

● MEDIUM

● **ADVANCED MATH**

Advanced Math

06

10 questions · ~15 min

Q1

ADVANCED MATH

$$(2x+5)^2 - (x-2) + 2(x+3)$$

Which of the following is equivalent to the expression above?

A. $4x^2 + 21x + 33$

B. $4x^2 + 21x + 29$

C. $4x^2 + x + 29$

D. $4x^2 + x + 33$

$$f(x) = (x + 6)(x + 5)(x - 4)$$

The function f is given. Which table of values represents $y = f(x) - 3$?

A.

x	y
-6	-9
-5	-8
4	1

B.

x	y
-6	-3
-5	-3
4	-3

C.

x	y
-6	-3
-5	-2
4	7

D.

x	y
-6	3
-5	3
4	3

Q3

GRID-IN

ADVANCED MATH

$$7x^2 - 20x - 32 = 0$$

What is the positive solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$(5x^3 - 3) - (-4x^3 + 8)$$

The given expression is equivalent to $bx^3 - 11$, where b is a constant. What is the value of b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

ADVANCED MATH

An object is kicked from a platform. The equation $h = -4.9t^2 + 7t + 9$ represents this situation, where h is the height of the object above the ground, in meters, t seconds after it is kicked. Which number represents the height, in meters, from which the object was kicked?

- A. 0
- B. 4.9
- C. 7
- D. 9

Q6

ADVANCED MATH

$$x^2 = -841$$

How many distinct real solutions does the given equation have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

Q7

ADVANCED MATH

Which of the following is equivalent to $\sqrt[4]{x^2+8x+16}$, where $x > 0$?

- A. $(x+4)^4$
- B. $(x+4)^2$
- C. $(x+4)$
- D. $(x+4)^{\frac{1}{2}}$

Q8

ADVANCED MATH

In the xy -plane, the y -coordinate of the y -intercept of the graph of the function f is c . Which of the following must be equal to c ?

- A. $f(0)$
- B. $f(1)$
- C. $f(2)$
- D. $f(3)$

Q9

ADVANCED MATH

$$5x^2 - 37x - 24 = 0$$

What is the positive solution to the given equation?

- A. $\frac{3}{5}$
- B. 3
- C. 8
- D. 37

Q10

ADVANCED MATH

Which expression is equivalent to $\frac{8xx - 7 - 3x - 7}{2x - 14}$, where $x > 7$?

- A. $\frac{x - 7}{5}$
- B. $\frac{8x - 3}{2}$
- C. $\frac{8x^2 - 3x - 14}{2x - 14}$
- D. $\frac{8x^2 - 3x - 77}{2x - 14}$